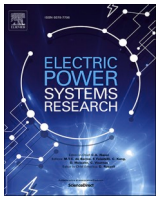
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Restoration of extra-high voltage power grids through synchronous and asynchronous hydro units during blackout—A comprehensive review and case study

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ABSTRACT

A blackout refers to a total power outage that rapidly plunges the electrical power system into a severe crisis, posing significant challenges for subsequent restoration efforts. Blackouts can occur due to a range of factors, such as delays in clearing faults, generator malfunctions, equipment failures, and human errors. After a blackout occurs, all possible measures are taken to achieve the fastest power system restoration. To ensure preparedness for such emergencies, scheduled black start exercises are conducted using units capable of initiating power restoration from scratch, allowing authorities to assess their readiness in catastrophic scenarios. Numerous similar exercises have been conducted regularly generating to highlight the complexities of power system restoration of the grid with the black starts of hydro generating units. This has aided plant operators in improving restoration methods and finding additional remedial steps to enhance existing processes to achieve an effective black start exercise. This paper overviews the multiple mock black start exercises applied to hydroelectric generating units and discusses several blackouts worldwide. In addition, Tehri Hydro Power Plant is considered for the case study. Tehri HPP is linked to the 400 kV EHV grid, which was later upgraded to 765 kV transmission lines. Further, this paper observes and discusses various challenges encountered during mock black start exercises. Possible remedial actions are suggested to address these challenges.

1. Introduction

With Several large-area blackouts have occurred in recent years in India, United States, Indonesia, Uruguay, China, Canada, Azerbaijan, Sudan, Kenya, Sri Lanka, Ukraine, Turkey, Pakistan, Bangladesh, Philippines, Thailand, Vietnam, Italy, Brazil, etc. [1]. The blackout in the northeast US and Canada on 14th August 2003 [2] and the tsunami-induced blackout occurred in Japan on 11th March 2011 [3]. On 30th July 2012, the blackout in India directly impacted the lives of almost 600 million people, making it the world's greatest power outage in history [4]. The 3.21 blackout in Brazil in March 2018 impacted nearly 53 million people, over one-fourth of the country's population, which resulted in a blackout load loss of 9300 MW [5]. Because Taiwan lies in the Pacific Ocean seismic zone and has a subtropical monsoon climate, earthquakes, landslides, typhoons, and mudflows threaten power system infrastructure. The increasing load has reduced Taiwan's Power System stability and increased the risk of blackouts [6]

Other causes of blackouts are fault clearance delay, voltage collapse, tripping of generating units and interconnected transformers, equipment failures, slow response of generating units, cyberattacks, human or operator errors, software failures, high temperatures etc. In addition, blackouts can also be triggered by various natural causes, such as adverse weather conditions like heavy rainfall, fog, dust storms, and natural disasters like hurricanes, thunderstorms, lightning strikes, blizzards, and high wind speeds, among others [1–5]. Table 1 lists recent blackouts and the damage these outages have caused.

In the context of power systems, a "black start" is the process of restarting the generator unit(s) without using off-site power for system restoration after a shutdown (partial or complete). Blackouts are rare, but they bring major economic and societal repercussions. Hence it is worthwhile to spend money and time getting the system ready for restoration using black start resources [32].

An efficient black-start strategy is critical to address the increasing risk of blackouts and enhance grid resilience. Restoration is the primary

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task immediately following a severe blackout [33]. Readiness for power system restoration of the grid with the generating unit capable of black start should be ensured at all times is crucial in the event of such emergencies. As a confidence-boosting exercise to assure success during the event of contingencies, regular mock drills of particular steps in the process of restoration should be performed. The key stages involved in the Mock exercise are as follows: (i) Initial switching operations to create subsystem with identified radial load, (ii) Testing of black start unit for islanding operation and checking its synchronising capability, (iii) Manually tripping the unit to create a blackout on the island, (iv) black starting of hydro generating unit (HGU), also known as black start unit (BSU), and (v) Lastly, confirm the synchronisation of the island with the rest of the electrical grid. [34].

Two significant methods for system restoration after a blackout are the top-down and bottom-up approaches [35]. The top-down approach is employed when the system has limited or no generating units with black start capabilities. In contrast, the bottom-up approach utilises a sufficient number of generating units with black start capabilities when neighbouring power grids are unavailable for assistance [36].

In 2021, the capacity of new renewable energy generation has reached 81 % compared to 79 % in 2020 [37]. Global generation capacity is 3064 GW at the end of 2021. With a 1230 GW capacity, hydropower contributed the largest percentage of global production, excluding pure pumped storage (Fig. 1). There was an additional 130 GW generation by pure pumped storage at the end of 2021 [38]. Therefore, the total hydropower capacity, including pure pumped storage, is 1360 GW (Fig. 2). In 2021, renewable energy capacity increased by 257 GW (+9.1 %). From Fig. 3, the hydropower capacity increased by 19 GW (+2 %). From 36.6 % in 2020 to 38.3 % in 2021, the renewable share of the overall generation capacity increased by approximately 2 % [38].

From the above discussion and Figs. 1–3, we can see that the renewable generation capacity is continuously increasing over the last decade. Moreover, over the past few years, renewable generation has been integrated on a massive scale into power systems. As a result, converter-based renewable generation displaces traditional synchronous generators like thermal units, lowering system inertia and frequency and voltage control capabilities [39–41]. Because conventional generation units are frequently offline to accept as much accessible renewable energy as feasible, the number of generating units willing to assist in restoration after a blackout at any particular moment may be restricted [35].

Cold-starting and getting such machines online might significantly lengthen the duration of a power outage. Apart from traditional

Table 1
Timeline of major blackouts events in the 21st century.

Date of occurrence	Location	Reason of blackout	Population affected (millions)	Time span	References
2nd January 2001	India	A failure at the UP substation	230	12 h	[7]
14th August 2003	USA, Canada	Software bug failed to alarm operators to redistribute the overloaded transmission lines	55	–	[8]
3rd September 2003	Italy	Cascading effect disrupted power from France & Switzerland due to trees falling on the load line	56	12 h	[9]
18th August 2005	Java-Bali, Indonesia	Cascading failure due to failure of 500 kV transmission line (power grid dissolve into two networks)	>100	3 h	[10]
29th August 2005	USA	Due to Hurricane (Katrina)	2.6	–	[11]
4th November 2006	Germany, France, Italy, Belgium, Spain & Portugal	Planned routine disconnection of the Ems powerline crossing in Northwest Germany to allow ship beneath the overhead cables	>15	2 h	[12]
8th September 2011	South-West, USA	Tripping of transmission line followed by cascading outages of the line leading to partial system collapse	>2.7	12 h	[13]
30th July 2012	Northern & Eastern India	An overdraw of power above the permitted limit caused a cascading failure	400	15 h	[14]
31st July 2012	Northern & Eastern India	An overdraw of power above the permitted limit caused a cascading failure	621	13 h	[15–16]
October 2012	USA	Due to a Hurricane	8	>1 week	[17]
21st May 2013	Southern Thailand	Failure in the HV transmission lines	7.9	11 h	[18]
22nd May 2013	Vietnam	The tree fell due to Crane Operator	10.2	9 h	[19]
6th August 2013	Philippines	The cluster of lines failed led to voltage collapse and caused the power outage	>8	13 h	[20]
1st November 2014	Bangladesh	Transmission lines failure	15	2 days	[20]
26th January 2015	Pakistan	The sudden drop in frequency & cascading effect	>140	3 h	[21]
31st March 2015	Turkey	A high transfer impedance resulted from all of the series capacitors being out of service.	71	5 h	[22]
21st November 2015	Ukraine	Saboteurs destroyed power lines	>2	5 h	[23]
23rd December 2015	Ukraine	Cyber-attack on Ukraine power grid	0.23	1–6 h	
13th March 2016	Sri Lanka	Technical problem	21	>8 h	[24]
7th June 2016	Kenya	The transformer tripped due toppling of an animal	10.1	5 h	
8th March 2017	Michigan, USA	Due to high-speed winds of up to 110 km/hr	1.1	> 7 days	[25]
20th September 2017	Puerto Rico, USA	Due to hurricanes namely, Maria & Irma	6.8	> 7 days	[26]
10th, 21st Jan & 27th Feb, 2018	Sudan	Trip sequences for generation bus bars operating without frequency load shedding have worked as intended.	41.49	24 h	[27]
21st March 2018	Brazil	An interruption in a transmission line near the HPP.	>10	1 h	[28]
3rd July 2018	Azerbaijan	Overheating at the Thermal Power Plant (fire started when cooling fans stopped working).	8	>7 h	[29]
20th December 2018	Lower Mainland, Vancouver & Gulf Island Canada	Due to high-speed winds of up to 100 km/hr	> 0.5	> 1 week	[30]
16th June 2019	Argentina, Uruguay & Paraguay	Operative error	48	4 h	
4th August 2019	Java	Disruption in the power line (high-voltage)	>100	20 h	[11]
14–17 & 17–18 February 2021	Texas, USA	Winter storms lead to energy infrastructure failure	>5	4 days	[31]

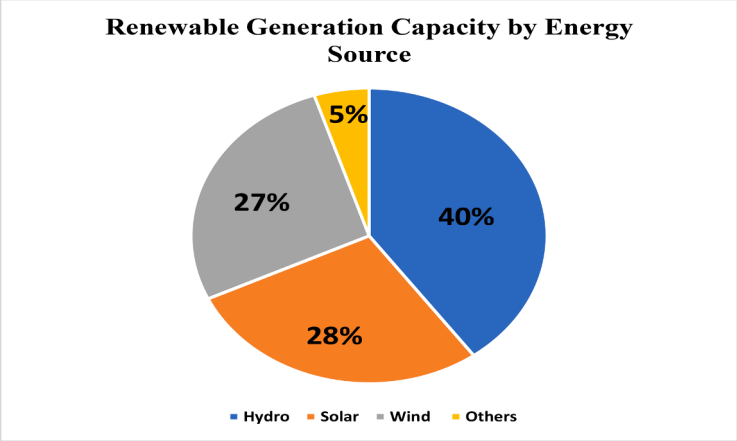


Fig. 1. Renewable generation capacity by energy source.

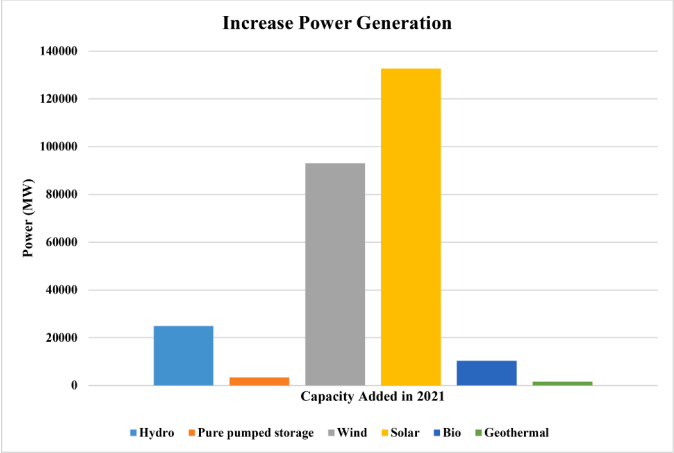


Fig. 2. Renewable power capacity growth.

generators like gas and hydro generating units, wind power generating units can contribute to the starting stages of restoration as they have more flexibility and controllability [42]. Due to the increased complexities and uncertainties, power systems are still in danger of catastrophic outages or blackouts [43].

Fixed-speed PSPP based on synchronous machines should be powered by power electronic converters that have a rating equal to the rating of the machine to enable variable speed operation (Fig. 4). This type of converter with high capacity is uneconomical [44]. Additionally, for underground power plants, employing full-scale converters with a rating of more than 200 MW poses significant challenges due to their large physical footprint, high expenses, and the need for extensive site clearance [45]. As a result, at sites with broad water head changes, variable speed PSPP using a Doubly Fed Induction Machine (DFIM) with partially rated power converters is an acceptable choice [46]. The DFIM has advantages in terms of electrical, mechanical, and economic perspectives [47]. The primary advantage of this strategy is that the power electronic converter, with a rating matching the slip power, will typically be a fraction of the machine's rating. The slip decides the rating of the converter, which makes the converter partially rated [48]. Another benefit of DFIM is that the power converters can be used to excite the stator from the rotor circuit, thus eliminating the requirement for an external VAR compensator [49].

Moreover, it provides better active and reactive power control and has a larger spinning reserve capacity [50]. That is why DFIM-based variable speed-based PSPPs are emerging worldwide [51]. For instance, there are notable examples of variable speed plants, such as the 800 MW facility in Ohkawachi, Japan; the 1000 MW PSPP in Linthal, Switzerland; and the 1000 MW PSPP in Tehri, India, utilising 3-level voltage source converters (currently under construction). Moreover, several projects in India are in the planning stage [52]. However, it should be noted that the DFIM lacks black start capability due to the absence of an auxiliary source to maintain the "pre-charge" state of the DC Link. The excitation control system of DFIM, comprising sensors, controllers, and converters, adds complexity [53]. Hence, research is essential to explore the black start capability of variable speed HGUs (DFIM), as shown in Fig. 5.

1.1. *India blackout: black start restoration and key details*

The power blackouts on July 30th and 31st, 2012, were unprecedented, affecting 350 million and later 670 million people, making them the most significant power crises in history, impacting one-tenth of the global population across 21 Indian states (Fig. 6) [15,54].The unfolding of events during the power outages in July 2012 closely mirrors the characteristics of the classic 'spiral breakdown effect' observed in power

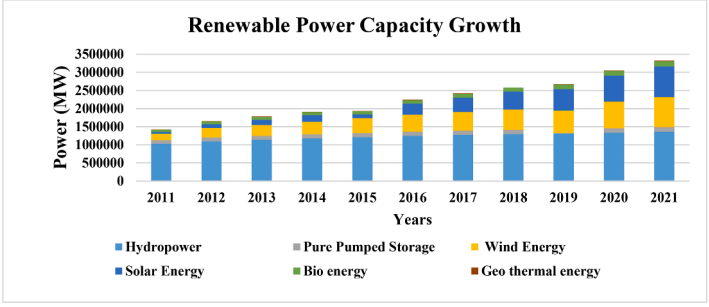


Fig. 3. Renewable power capacity growth.

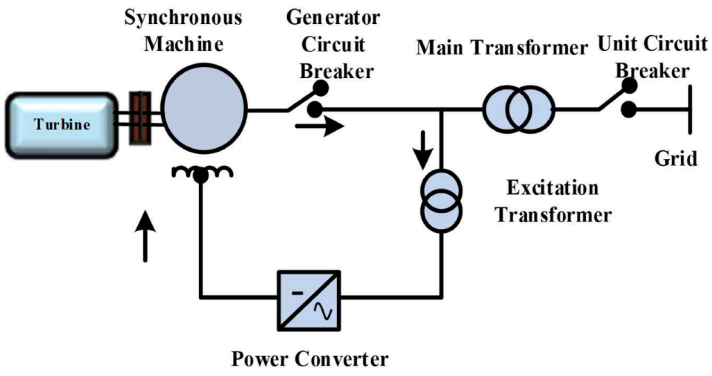


Fig. 4. Synchronous machine based fixed speed hydro generating unit.

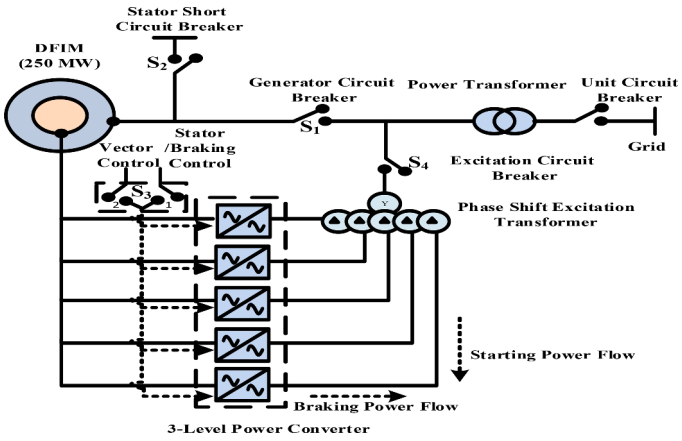


Fig. 5. Doubly fed induction machine based variable speed hydro generating unit.

systems [16].

1.1.1. Causes of outage

Reports from India's regulatory authority, the Central Electricity Regulatory Commission (CERC), underscore that the NEW grid faced substantial stress throughout July due to a complex interplay of factors [16,55]. The absence of monsoon rains led to an unprecedented surge in agricultural demand in the Northern region, aggravated by an excess of power in the Western region. Furthermore, only 2/4 critical 400 kV interconnections between the West and North were operational, exacerbating grid vulnerabilities [4].

The critical turning point occurred when one line was deliberately taken offline for planned upgrades, and the other experienced an unexpected equipment failure [56]. Alarmingly, the pivotal Agra-Gwalior line neared collapse 36 h before the major grid breakdown, yet no significant measures were taken to manage demand or curtail generation. Instead, surplus power from the West was redirected through central and eastern states to the power-deficient North [57].

In this precarious situation, state-owned utilities neglected Under Frequency Relays (UFRs) on their power lines, often due to political pressures. These UFRs, designed to respond to frequency drops, went unactivated, leaving the system vulnerable. Tragically, this led to a cascading series of shutdowns, culminating in the historic power outages. Fig. 7 illustrates events starting at 12:10 AM on July 30th, with the West–North connection weakening, isolating the Northern region from the NEW Grid, resulting in a widespread blackout [16].

Within 32 h of the initial outage, a second disturbance in the NEW grid occurred at the West-North interface. The circumstances mirrored

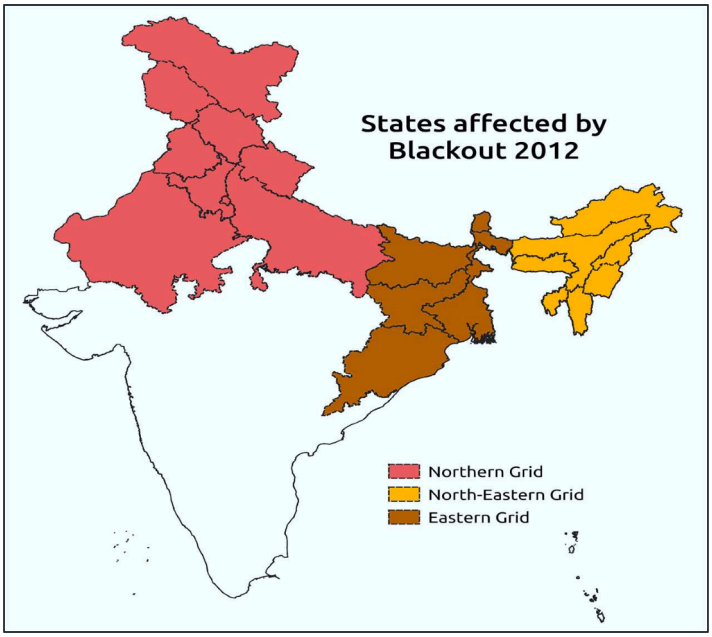


Fig. 6. India's map highlighting the states where the power goes off during blackout 2012 [15,16].



Fig. 7. Sequence of events leading to the first blackout (30th July 2012) [15,16].

the previous day, with heightened power demand in the North and surplus power in the West. Despite efforts to restore supply, two NR-WR 400 kV links failed at 1:00 PM on July 31st, triggering a chain reaction of shutdowns across other NR–WR links. With many WR-ER connecting lines still inactive, the system was vulnerable [58]. Consequently, 38 links disconnected rapidly due to excessive load, causing a substantial reduction in generation capacity [4]. This severed the Northern, Eastern, and North-Eastern regions from the Western grid, resulting in a widespread blackout. Meanwhile, swift isolation and generation unit tripping prevented an outage in the Western region (Fig. 8).

1.1.2. Restoration of supply

Efforts to restore power supply began promptly on both days, but due to the extensive disruption, full grid functionality was a time-consuming endeavor. The gradual process of bringing generators back online, synchronized with specific load requirements, led to a prolonged recovery, with some regions experiencing up to 16–20 h of power loss [59].

The generation mix complexity was a contributing factor. Delayed monsoons reduced hydroelectric power, increasing reliance on slower-starting thermal plants (coal), which required 6–8 h for recovery.

Critical loads, like railway traction and major hospitals, were prioritized for quick restoration. New Delhi Airport restored power within 15 s due to robust backup systems [60]. Businesses invested in backup solutions, including diesel generators, to cope with intermittent outages. Industrial facilities relied on Captive Power Plants to mitigate production disruption.

CESC (Calcutta Electricity Supply Corporation) and DVC (Damodar Valley Corporation) Role in supplying power during Blackout 2012:

CESC: During the triple grid failure in North India, the East, and Northeast, Kolkata was spared disruptions due to a strategic action by utility provider CESC, led by Vice-Chairman Sanjiv Goenka, who received a critical call alerting him to a potential grid issue [61]. He promptly disconnected CESC from the grid, crucial for Kolkata's resilience. Sensors at Howrah junction aided the action, and CESC's full capacity operation supported state grid restoration, maintaining uninterrupted essential services.

DVC: It emerged from a century-long effort to control the volatile Damodar River, triggered by the catastrophic 1943 flood and championed by Dr. B.R. Ambedkar [62]. The committee recommended a TVA-inspired authority, and DVC, India's first multipurpose river valley project, was established in March 1948 by the DVC Act, led by Mr. Voorduin's comprehensive plan. [57].DVC plays a crucial role by supplying bulk power, including 11 KV, 25 KV, 33 KV, 132 KV, and 220 KV [63], to vital sectors like railways, steel, etc., supporting our national economy. Important Milestones of DVC are given in the Fig.9.

2. Overview of the restoration process

At the time of a power plant outage, operators can depend upon the electric grid to which the plant is connected to give power supply to house plant loads. Also, essential auxiliaries can start generating units or keep them running (such as gas compressors, oil pumps, draft fans, pulverisers, controls, switchgear and sensors etc.). But in large outages event when the grid itself gets failed. Thus, unable to give the power needed to any power generation plants within its service territory. Thus, the requirement for a plant capable of starting its generator(s) without using external power from the grid arises. This process is called "Black Start" [53]. The following properties need to be checked when assessing whether a generating unit can be a black start resource: small station power demands, fast restart, high ramp rate, size, onsite supply of fuel should be sufficient, stabilisation of system frequency, inertia and location.

A typical black start restoration consists of the following three stages:

2.1. System preparation

The preliminary preparations need to be done before the start of black start restoration to avoid any interruptions. The main motive of this stage is to know the power system status and the required method for restoration. A preliminary examination should be done to know whether the condition of restoration exists. Then, identification of resources that are available for use [64].

Before the start of the restoration process, the plant operator needs to check generating units,tie-lines and neighbouring systems. They also

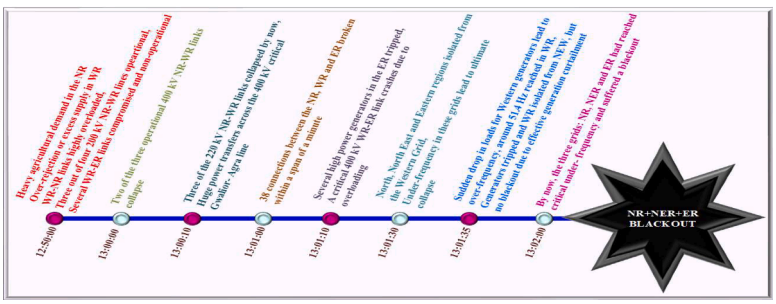


Fig. 8. Sequence of events leading to the second blackout (31st July 2912) [15,16].

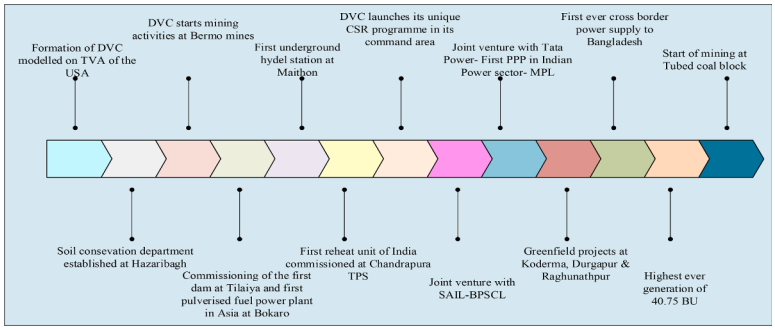


Fig. 9. Important milestones of Damodar Valley Corporation (DVC) [15].

need to check whether any portion of the system is in operation state or islanded mode. Moreover, it would be better to have some information that can help to know the disturbance cause that may help us avoid recurrences at the restoration times as not re-energising faulted part of the system [65].

2.2. System restoration

As the plant operators are done with preliminary preparations, the restoration process of the system can be started. During this phase, black start resources can be utilised for the energisation of cranking paths, and power for starting-up is given to generating units that can assist in restoration. Close coordination and communication are needed to ensure all involved actions for restoration take place on time and avoid any pitfalls [66] Power system operation during the restoration process faced many challenges, including.

(i) Voltage Control: Runaway voltage conditions can be caused while charging transmission lines and reactive devices operation, which leads to equipment breakdown or system failure [67]. Plant operators should pay extra attention while voltage control during restoration and ensure it lies in defined operating parameters [68].
(ii) Frequency Control: To maintain frequency at the time of restoration is very difficult as any changes in the generation side or load side have more impact on the small system. Charging a single circuit can lead to deviations, and cold load pickup can affect frequency [69].
(iii) Equipment Issues: Due to Initial disturbance, some equipment may get damaged while some are not operating due to stored energy loss, failure of control systems, etc. Protective relaying problems can also arise due to reduced fault current and abnormality in system configurations. That's why various schemes for protection are not proper for restoration. Backup systems operations and equipment need to send off promptly to remote locations to avoid restoration delays [70].
(iv) Problems with Stability and System Dynamics: Throughout the restoration phase, the power system is more susceptible to problems with system dynamics. During the operation of a weak system, maintaining dynamic stability is a challenging task, and care is needed to limit transfer across transmission lines voltage control to avoid stability problems [71].

2.3. Load restoration

The critical load needs to be identified for making successful restoration. Critical loads are required for the restoration process and are important for the sustained operation of the bulk electric system [72]. The plant operator should first identify the load that needs to be energised to speed up power system restoration (PSR). Loads crucial for substations and that support infrastructure have more priority. Other loads need to be picked up for frequency and voltage control.

Traditionally, hospitals' life-supporting medical equipment is prioritised for restoration even if they have onsite power backup. It is advised not to pick up large loads so early at the start of the restoration process. Limiting load to less than 5 % of total synchronised generation is recommended. The load can be restored more rapidly as the restoration process goes on as non-black start generation starts recovering. Due to its flexible generation, hydro is opted for a considerable amount of generation reserves prior to adding large loads [73,74].

The PSR process will end as soon as the system is in a situation when the addition of the next load is independent of frequency or control needs. Finally, the power system operations are back to normal. Integrating renewable energy sources (RES) into the power system also leads to a renewable power solution for PSR [75]. The black start process using renewable energy resources and a battery energy storage system is shown in Fig. 10.

3. Methods of black start adopted worldwide

Various black start methods have been adopted worldwide and have gained the attention of researchers in the mid-nineties. Initially, they used combustion turbines for the black start process and then shifted to hydro turbines using diesel generators (DG) as a backup supply. The research is moving towards replacing DG sets due to their various drawbacks with RES as the backup power source for essential auxiliaries during black start. But renewable resources have their own limitations as they are not always available, which leads to the requirement of an energy storage system [76]. Based on the approaches adopted, the methods are broadly classified as (i) Conventional Approaches and (ii) Modern Approaches.

3.1. Conventional approaches

It started with black starting generating stations from hydropower units and combustion turbines at a remote location. Simply activating a valve that allows water to flow into the generator allows HGU to create station power. The properties of gas turbines include quick starting and ramping [53]. The field tests were done on actual systems and then compared results to the analytical analysis [77]. Combustion turbines have been installed at remote locations from generating stations due to various physical limitations or operational considerations [78].

The restoration of generating units depends upon the backup supply to light up essential auxiliaries during black start. In the event of a plant shutdown, this supply may be provided by a neighbouring power plant [79]. But in case of a widespread outage when the electric grid is also out of service, the thermal capacity of boilers is sufficient enough to light up auxiliaries or the main unit [80]. So, the steam-driven dc sets can provide power for emergency lighting. Further, a knowledge-based system was used to automatically control a power system during restoration [81]. But the problem-solving ability of the human mind was superior to the computer systems of that time.

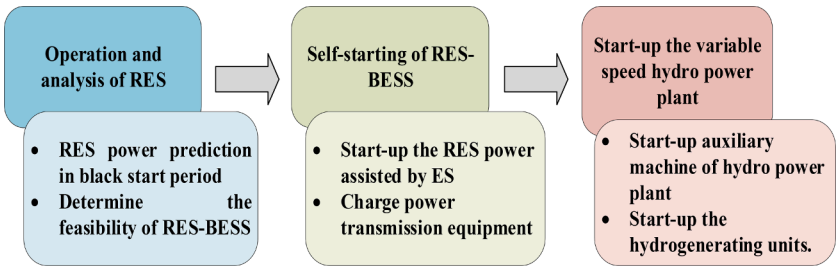


Fig. 10. The black start processes of power grid using renewable energy resources and battery energy storage system (RES-BESS).

Then at a later stage, a method that uses hydro generation at a remote location to restore coal-fired generating stations [82]. Coal-fired stations induce draft fan, start-up boiler feed pump, reservoir & circulation water pump motor, etc. These electric motors are considered important loads to bring back each unit online. But at the same time, these large motors face problems during starting. These motors draw high currents and require more starting time. The order of starting motors is determined by size, first the largest and then in descending order. The guidelines for restoration have been set, beginning with selective hydropower plants and slowly reaching the thermal ones [83]. Different electricity boards periodically carried out field tests to improve system restoration reliability. Black start resources (BSR) start the system restoration process and recovery of loads. BSRs have onsite gas or DGs that can be started and deliver power to the generating unit

Table 2
Black start methods adopted worldwide (Conventional approaches).

Researcher (s)	Year	Methods adopted	Black start resource	Country	System	Findings
H.A. Dryar et al. [69]	Oct 1940	By starting each individual station separately as generating capacities & load of each major distributing center approaches a balance	Thermal capacity of boilers	Philadelphia, USA	Transmission system, transformers, loads, hydro & steam generators	Sectionalised ring buses will be preferred over double buses in future
R.J. Kafka et al. [84]	Aug 1981	Operating personnel & system analysts works in close coordination, data is accessible through online and use load flow program	Drum type boiler	Washington, DC	HV transmission lines, no of large n/w loads, & generators	Corrective measures prevent system problems from being fully collapsed by stopping them. Combined efforts of system analysts, system operators, & production personnel are needed
T. Sakaguchi et al. [75]	Feb 1983	Development of knowledge-based system for PSR	–	Hyogo, Japan	Transmission lines, sources, loads, and generators.	Heuristic program approach with KBS Lisp program runs slowly on medium size computer Lisp machines address this in the future.
R.D. Schultz et al. [85]	Aug 1984	Black start a generating unit from a combustion turbine generator placed at remote location using field test & analytical analysis	Combustion turbine generator	Potsdam New York	28 MVA Combustion turbine generator, generating unit, 138 kV transmission line, transformer and load	Highlight the significance of defining the motor insulation's voltage withstand level. It is advised to use the standard turn-to-turn insulation specification and the OCMA ground wall. Surge comparison & surge testing to identify weakness b/w turn-to-turn insulation
J.H. Jones [86]	June 1989	A diesel generator is provided to supply emergency supply rated 200 kW at 480 V. Its controls are configured to auto-start and load upon loss of normal station service supply.	Diesel engine generator		Two 480 V load centers, each fed from 750 kVA, 6900/480 V transformer.	Engine generator was sized to supply one governor oil pump, the power house sump pumps, one battery charger, one speed increaser oil pump, emergency power house lightning, fire pump and service water pump.
Randy R. Lindstrom [87]	Feb 1990	Remote hydro generation to restart a major coal-fired generating station using field tests and analytical study	Hydro generator (Francis & Kaplan)	Nebraska, United States	Motors, hydro-generators, and transmission system	Black start expectations for GGS are realistic and achievable Minor system modifications & procedural changes are needed Field Tests & analytical study correlated
P.G. Boliaris et al. [88]	April 1994	Simulation of energisation of transmission line for BS studies applying EMTP.	Hydro power station or gas turbine	Athens, Greece	Three-phase synchronous generator, and transmission lines (distributed & lumped parameters line model)	Surge arrester reduces over voltages to end where it is connected Dynamic behaviour doesn't affect developed over voltages This phenomenon affects the field current
B. Delfino et al. [67]	Aug 1996	Field tests taking adv. of maintenance outages of thermal groups	Hydro & thermal units	Italy	Italian HV n/w near French border	Black start has been improved by field tests outcomes Field test data availability helps in simulation studies
Nitish Saraf et al. [76].	Nov 2009	Annual black start service selection analysis through competitive process (price-based bids)	Not specific	Texas,US	ERCOT grid	Uses conservative method Low flexibility in changing search criteria Assure ERCOT contracted services is optimum solution

auxiliaries [76]. The various conventional methods of black start adopted worldwide are tabulated in Table 2.

3.2. Modern approaches

A sectionalising method was used for power system restoration PSR more than ten years ago. This method does restoration in a separate part, i.e. islands in the power system, and later interconnects them. Various algorithms have also been developed considering different parameters such as the black start capability of generators, power supply-demand balance and independence of Islands. Moreover, the islands can be easily observed using the Wide Area Measurement System (WAMS) [89]. It is essential to know the availability of BSU to supply power (cranking) to non-black start units (NBSU) to maximise the generation capability of the entire system [90].

But many time-changing practical constraints need to be checked for optimisation. Mixed-integer linear programming (MILP) has been used for the sequencing of generator start-up. This linear programming gives optimal solutions better than heuristic or enumerative techniques. This method can also provide the system with MW generation capability as the PSR process continues [65]. Further, a decentralised multi-agent system (MAS) approach is developed for PSR that considers load demand uncertainty and renewable distributed generators (RDGs) [91]. In this load, demand uncertainty and generation of RDG are calculated by the use of Monte Carlo simulation to create scenarios. Then the node in the island to be restored is found using a rule-based heuristic technique [92]. The maximum likelihood estimation technique is used to find the maximum likelihood of the island being restored [93].

DGs are commonly used in HPPs to light up auxiliaries and supply essential auxiliaries during the black start. However, DGs come with numerous drawbacks, particularly concerning the environment, including high costs, frequent maintenance requirements, noise pollution, susceptibility to corrosion, and the need for winterising the generator in cold weather. Additionally, DGs can be excessively expensive and are often under-utilised as backup power sources. [94]. Hence renewable energy solution is needed, but it has their limitations [95]. As RESs are not always available, an energy storage device is also required (Fig. 11) [74]. But the ES device has a charge/discharge power and capacity restriction due to the volatile nature of RES [96]. The asynchronous machines are mainly used in wind and hydro to create more opportunities and challenges and control them to create a stabilised RES [97].

Integrating RES in the power system demands wind farms' involvement in the PSR. Black Start needs wind farms to control voltage and frequency. Grid forming control for DFIG is used for restoration [98] as shown in Fig 12. The rapid increase in distributed inverter-based resources (IBRs) brought researchers' attention to whether IBRs can be used to black start microgrids and distribution feeders [99,100]. The black start option is also necessary for pump storage hydropower stations for variable speed operation. The DFIM is already being used in wind power plants. But it is a new addition to variable-speed hydropower energy conversion that requires extensive research on electric power system restoration. The various modern methods of black start adopted worldwide are tabulated in Table 3.

4. Mock black start exercise: a case study at Tehri HPP

The restoration process needs to be carried out when a blackout occurs. The operators must play their respective roles without interruptions during the disturbance. Mock black start exercise needs to be performed regularly to build confidence and ensure success during blackouts or plant outages. It is recommended that these exercises be performed in locations where a unit is starting up or when power is being extended to nearby substations [106]. The Mock black start exercises may differ from the real-time black start exercises. India's Northern Region has organised various black start mock drills from hydropower plants available at Bhakra (BBMB), Chamera, Bairasiul, Tehri, Jhakri, Salal, and other locations (Table 4).

Black start mock drills need good coordination between load dispatch centres, substations, generating units, and the stations that are part of the exercise. A load (radial) nearby to generating station is to be determined in the planning stage. The switching diagrams are checked during different stages to determine circuit breaker positions. The plant operators should check the already available switching settings, protection settings, and synchronising facilities. The responsibilities and roles of operators participating in the exercise and comprehensive procedure telling different steps are developed. All the distribution and transmission substations participating in the black start exercise must be informed to make sure that supply from multiple sources is not mixed due to unintentional switching operations [107]. Additionally, this is done to prevent the need for an alternative supply during a planned (simulated) blackout.

The preparatory steps, like splitting busses and load shifting, are completed during the execution stage. Then mock black start exercise is performed, and the system's parameters are noted. After the mock drill is completed, the experience is documented. The success and shortcomings during the exercise are reported to the concerned utilities. The next section explains the detailed mock black start exercise in Tehri HEP.

As shown in Fig. 13 following switching operations (SO) are suggested at 400/220 kV Meerut Power Grid, 400 kV Koteshwar pooling substation, 400 kV Tehri HEP, and 220/132 kV Shatabdinagar substation (UPPTCL) to support the Tehri Power House's mock black start exercise (UPPTCL) [108]. The supply at 220/132 kV Shatabdinagar (UPPTCL) would be interrupted during the mock drill. The procedure for Tehri HEP mock black start exercise consists of the following steps:

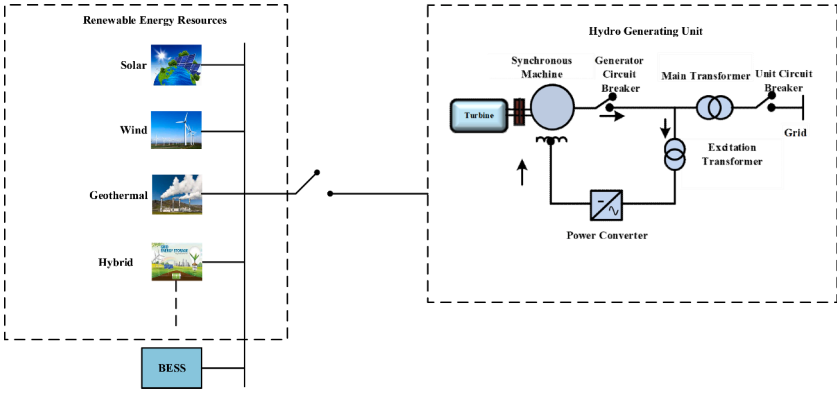


Fig. 11. Structural of black start system with renewable energy resources (RES) and battery energy storage system (BESS) as main power supply. The RES–BESS as black-start power to start auxiliaries of synchronous based hydro generating unit.

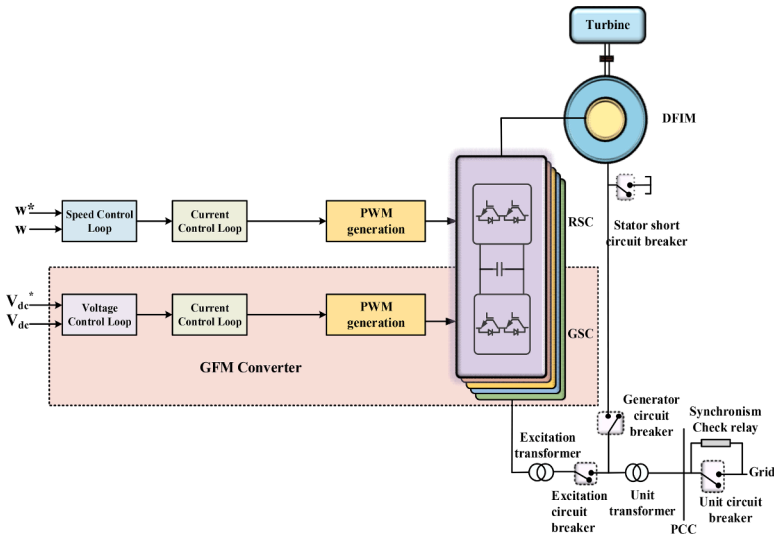


Fig. 12. Block diagram representing control structure of a grid-forming (GFM) converter for black start.

4.1. Preparation

Before the start of the mock black start exercise, some preliminary preparations need to be done that involve [109]:

(i) Ensure the healthiness of the DG set.
(ii) It is required to check whether the operator can start one hydro-generating unit using an auxiliary power source from the DG set.
(iii) It is also checked whether charging a dead 400 kV bus with BSU is possible.
(iv) It is required to check the necessary settings of the excitation system. So that 400 kV bus voltages can be varied smoothly throughout the full range of the generator capability curve.
(v) Make sure that synchronising trolley is in proper working condition.
(vi) Manpower to be assigned with due care for successful exercise and witnessing it for learning.

4.2. Initial switching operations

The Mock black start exercise is performed in close coordination with SLDC & NRLDC, UPPTCL (Lucknow), Tehri HEP, CPCC NR-1, 400 kV Koteshwar pooling station (PG) & 400/220 kV Meerut (PG), 220 kV Shatabdinagar Substation [110]. The goal is to perform SOs in order to build a subsystem with an identified load for the 220 kV Shatabdinagar (UPPTCL) substation while using the operational code from NRLDC to begin the SO for a mock exercise. SLDC and UPPTCL are responsible for checking that the loading is under control. 30–40 MW is the desired load on the island. Unit#1 has been presumed as the black start unit (BSU). However, Tehri HEP will decide which hydro-generating unit to be considered as BSU during exercise. It is to be checked that the isolators are opened after opening up the breaker to maintain voltage within the desired range [71].

4.3. Islanding scheme

Through a 400 kV bus coupler at Tehri HEP, the subsystem with 220 kV Shatabdinagar load and Unit#1 (designed for black-start) is connected to the grid. It is to regulate the generation of BSU at Tehri HEP to keep power flow at a minimum through the 400 kV bus coupler. The BSU generation is to be maintained as per the actual loading of the planned island in coordination with UPPTCL [111]. Let the subsystem be operated in the island mode for about 15 min by opening the bus coupler at Tehri HEP. The HGU at Tehri HEP in the subsystem should be on free governor mode, and AVR to be in auto mode. The frequency of the subsystem should be regulated, and it is to be operated for 5 min each at 50 Hz and 50.5 Hz. Synchronise the subsystem with the grid now without interrupting the load at Shatabdinagar. Trip/desynchronise BSU breaker [112].

4.4. Mock black start exercise

(i) Creating blackout at 220 kV Shatabdinagar (UPPTCL).

At Tehri HEP, open the bus coupler. The island would be re-created as a result of this. Trip BSU breaker. By doing so, the power supply to the island would be interrupted. Open breaker of Koteshwar pooling-Tehri ckt-1 [113]. Open breaker of Koteshwar pooling-Meerut ckt-1. Open tiebreaker of Meerut-Koteshwar pooling ckt-1 & ICT#1. Open 220 kV side breaker of ICT-1. Open breaker of Meerut–Shatabdinagar line. Following a voltage drop in the subsystem formed with the load. Open the 220 kV Meerut (PG) Shatabdinagar line breaker, the 220/132 kV transformers 1&2, the 132/33 kV transformers 1&2, the 33 kV feeders, and the 132 kV feeders that are kept in service during the exercise [114].

(ii) Black-Starting of Unit at Tehri HEP

Now BSU can be self-started at Tehri HPP. The plant operator should make sure that the auxiliary supply is brought from the dedicated DG set at the time of the real-time black start exercise [115]. Dead 400 kV bus-1 charges through BSU at Tehri HEP. The black start unit would be in free governor mode. Vary the excitation to keep the bus voltage at about 380 kV. So, keep the AVR set point accordingly. AVR should preferably be kept in auto mode.400 Tehri–Koteshwar pooling ckt-1 is charged once a clearance from the Koteshwar pooling substation is taken [79]. Close breaker of Tehri–Koteshwar pooling ckt-1. Close breaker of Koteshwar pooling-Meerut ckt-1. Line voltage needs to be observed on 400 kV Koteshwar pooling-Meerut ckt-1. The Voc should not exceed 410 kV. If there is a high voltage, inform engineers at Tehri HEP to regulate excitation to reduce the open-end voltage at 400 kV Meerut to the desired level. By this, it can be ensured that over-voltage conditions will not persist when the tiebreaker closes. If the voltage is as per desired limits, the tie breaker of 400 kV Koteshwar pooling–Meerut ckt-1closes.By doing so the 400/220 kV ICT-1 at Meerut and 220 kV bus-1 at Meerut energise. Main breakers of Koteshwar pooling-Meerut ckt-1 & ICT-1 are to be kept open. Close breaker of the 220 kV Meerut-Shatabdinagar line [116]. Keep the 220-kV bus coupler open. Charge 220 kV bus at Shatabdinagar by closing the breaker of the Meerut-Shatabdinagar line after

Table 3
Black start methods adopted worldwide (Modern approaches).

Researcher(s)	Year	Methods adopted	Black start resource	Country	System	Findings
S. Arash Nezam Sarmadi, et al. [89]	Aug 2011	Sectionalising method	Generating set	Iran	39 bus & 118 bus power system	dividing a minimally built PMU network into individually restorable islands Fast & secure Provides with phase angle diff between adjacent stations using WAMS
Wei Sun et al. [65]	Aug 2011	generator start-up sequencing strategy as a MILP	Generating set	United States	IEEE 39-bus system, AEP, & energy tests cases	Numerical Results validate 1.Accuracy of models 2.Computational η of MILP
Anurag Sharma et al. [92]	July 2018	Decentralises multi-agent system (DMAS) approach	EVs & BES	Singapore	38 & 119 bus distribution system	Development of controlled islanding Implementation of DMAS architecture Scalable & robust EVs & BES impact
Georgios Patsakis et al. [101]	Nov 2018	Randomised heuristic approach to solve optimisation problem to optimally allocate BSUs	_	USA	IEEE-39, IEEE-118, and WECC system with 225 nodes & 136 gen	Formulated and solved optimal BS allocation problem Proposed heuristic by LP relaxation of optimisation
Roberto Benato et al. [36]	April 2019	Robust model with diff. grid configuration	Generating set	Italy	Model with single generating plant control	TERNA developed simulation tool in RME & EMT environment. Realisation & validation of dynamic models
System Restoration Procedure for Northern Region by NRLDC [102]	Feb 2019	Using synchronous based hydro generating unit	Diesel Generator set	Northern Region, India	BSU (synchronous generator), 765 kV line (400 kV prev) & loading station : actual system in Tehri HPP	Subsystem frequency: 50.00–50.50 Hz. Radial load to be fed by x`mers at loading station Load to be inc gradually in step of 5 MW Desired line & gen voltage should be maintained
Lei Sun et al. [64]	May 2019	Comprehensive Skeleton-network restoration (CSNR) scheme	Generating set	China	10 unit 39-bus and an actual power system in Guangdong, China	CSNR is proposed for PSR planning GSUs optimises generator start-up sequence TLR optimises restoration sequence of TL LDP model calculates the quantities & locations of local loads
Junhui et al. [96,103]	Aug 2019	Stratified Optimisation Strategy	PV-BESS	China	Optimisation coordination layer, of PV-BESS, control strategy of PV and ES	PV power index / prediction & optimisation layer designed to meet PV output power requirements. Design to solve the problem of uncertainty of PV-BESS & over -charging/discharging of ES
Jingxin Hu et al. [104]	10th Oct 2021	Closed-loop black start-up control using DAB converter		China	Downscaled DAB converter	Introduces black start-up control of DAB converter Start-up time is reduced up to 55.6 % to SOA
José Luis Rodríguez-Amenedo et al. [98]	26th Oct 2021	Grid- Forming Control Based on the Rotor Flux Orientation (DFIG)	Wind energy	Spain	Wind farm consisting of two DFIG wind turbines (WT) connected to PCC	Ability of DFIG based wind turbine to PSR GFM control based on rotor flux orientation Proposes WT control measure to supply isolated load
Abhishek Banerjee et al. [105]	July 2022	autonomous microgrid restoration idea by making use of grid-forming (GFM) inverter-based resources (IBRs) & smart CBs	IBRs	New Jersey, USA	Modified IEEE-9 bus system and custom 5- inverter (GFM) microgrid break into small MGs with smart breakers	Black started using inverters (GFM) & recombined with smart breakers Demonstration of two MG system restoration & power sharing performance 12 GFM inverters, 1 smart breaker, & loads for feasibility evaluation

line voltage is available. 220/132 kV transformers 1 and 2 charge that energises 132 kV bus. 132/33 kV transformer 1 &2 charges which will energise the 33-kV bus at Shatabdi Nagar [34].

Gradually add the 33 kV radial load in the slot of 5 MW. Close all 132 kV lines that are in use during the exercise. The Shatabdinagar substation will ensure that the 220 kV Modipuram and 220 kV Simbholi should never be closed [125]. The islanded operation should be observed for about 10–15 min.

Table 4
Details of black start capability of hydro generating units in northern region.

	THDC		NTPC	NHPC				BBMB	AD Hydro Power Ltd.	Jaiprakash Power Venture Ltd.	Punjab
Name of Generating Station	Tehri	Koteshwar	Koldam HEP	Chamera HPS-I	URI-I	Dulhasti	Parbati-3	Bhakra	AD Hydro	Karcham Wangtoo	Ranjit Sagar
Fuel Type	Hydro	Hydro	Hydro	Hydro	Hydro	Hydro	Hydro	Hydro	Hydro	Hydro	Hydro
Black start capability	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes
Self-start facility	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes
Installed Capacity (MW)	4 × 250	4 × 100	4 × 200	3 × 180	4 × 120	3 × 130	4 × 130	5 × 157	2 × 96	4 × 250	4 × 150
Black Start Source	DG Set	DG Set	DG Set	DG Set	DG Set	DG Set	DG Set	DG Set	DG Set	DG Set	DG Set
Capacity of DG Set (kVA)	2 × 1000	2 × 1010	2 × 1250	1 × 1010 & 2 × 1000	2 × 1000	2 × 640	2 × 1010	500	750	2 × 1500	2 × 500
Last date on which mock drill carried out	21-Mar-18	20-Dec-18	11-Jan-19	3-Dec-18	20-Dec-16	—	19-Dec-18	11-Dec-15	3-Jan-19	16-Jan-18	—
Self-started on 30/07/2012	Yes	No	Not	Yes	Yes	No	Not	No	No	Yes	No
Self-started on 31/07/2012	Yes	No	Applicable	Yes	Yes	No	Applicable	No	No	Yes	No

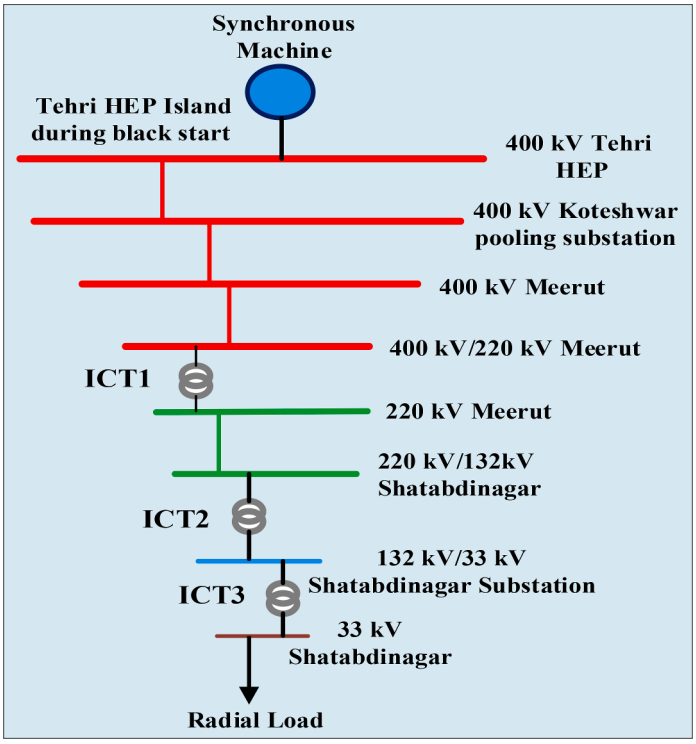


Fig. 13. Simplified scheme of the island grid during mock black start exercise at Tehri HEP.

4.5. Synchronisation and restoration

Synchronise the island system with the Meerut grid (400 kV) after taking code from NRLDC. The synchronisation can be done by closing the 400 kV Koteshwar pooling-Meerut ckt-1 or main breaker of ICT-1 or by bus coupler of 220 kV bus [117]. BSU at Tehri HEP would perform frequency matching. The plant operator at the Meerut end would make sure that closing occurs when the frequency and voltage of the subsystem and grid are within specified synchronisation limits. Meerut (PG) substation, Tehri HEP, and Shatabdinagar Substation should work in close coordination to regulate the generation, load, and synchronisation. BSU changes to power mode. 400 kV bus coupler of bus bar closed at Tehri end. DG stopped, and 11 kV incomer restored. Finally, restore the system to normal operation after synchronisation [101].

The EHV transmission lines are preferred because they have highly active and reactive power transfer capability. FACTS devices are commonly used to increase the power transfer capacity of EHV lines.

The transmission lines (400 kV) are not adequate to provide the active power of 1400 MW (rated), excluding pumped storage units of the system (Fig. 14). The 400 kV EHV lines using 35–40 % compensation (series) may enhance the line transfer capacity to 1400 MW at 0.9 pf. But the 765 kV EHV lines do not need any compensation (series or shunt). Also, it is adequate to provide 1400 MW at all ranges of power factors required and at the same time, it maintains grid stability. Also, 765 kV EHV transmission lines can provide power at the rated level and maintain the stability of the grid in case of a short circuit fault. Moreover, the present black start technique using one synchronous-based hydro-generating unit is insufficient to provide the reactive power required for

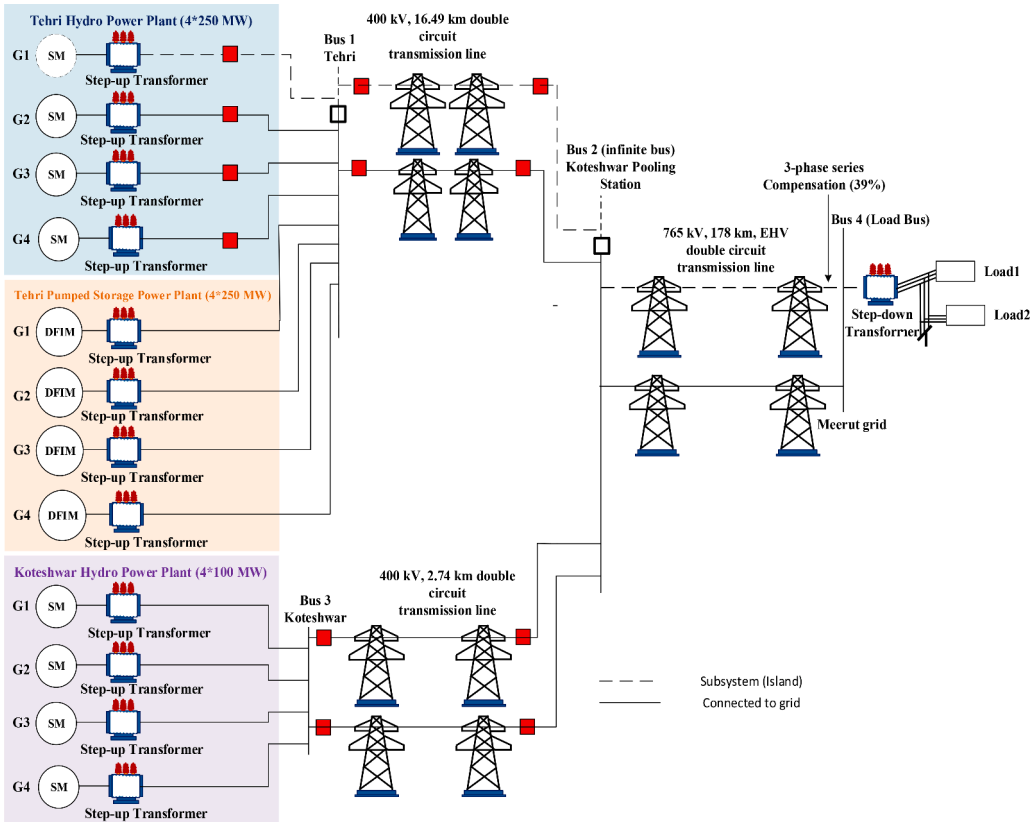


Fig. 14. Black start capability of hydro power plant connected to extra high voltage line (considering Tehri Hydro Power Complex).

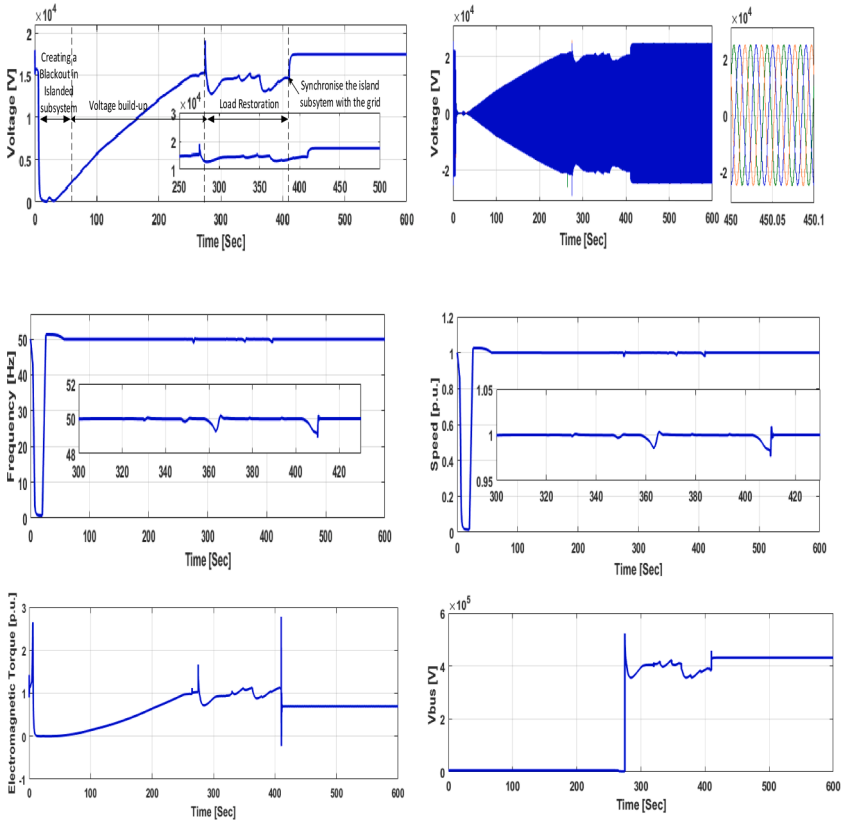


Fig. 15. Black start operation of fixed speed hydro generating unit (a) Voltage, (b) Three phase voltage (c) Frequency, (d) Speed, (e) Torque (Electromagnetic), and (f) Bus 4 voltage.

the 765-kV transmission line. Thus, research is needed to find the black start solution of the Tehri HEP with an upgraded 765 kV EHV from a 400 kV EHV transmission line [118].

Results of case study carried out at Tehri HPP are shown in Figs. 15 and 16.

5. Challenges and future scopes

By considering various aspects of major steps of mock black start exercises of the hydro generating units. Based on the discussion in the previous sections, this section provides some key points/ future research to be done so that the black start capability of the hydro-generating units can be enhanced [87].

5.1. Synchronous-based HGU

(i) ***The inrush of harmonic-rich current:*** During the black start, the energisation of transformers may lead to the inrush of harmonic-rich current. That current may resonate with an impedance of the system to produce overvoltages temporarily. So, the over-voltage protection setting needs to be taken care of during the black start process [34]. While charging ICTs, increasing the time delay setting of overvoltage stage protection is advisable to avoid tripping due to transient overvoltage. Further, it is recommended to charge EHV lines and ICT at a reduced level of voltage. Then it is gradually increased rated values using an automatic voltage regulator (AVR) [109,119].

(ii) ***Repeated tripping of the feeder (EHV)***: During the energisation of transformers, transient overvoltage emerges. When the lines are charging, the transient overvoltage rises to the range of over-voltage stage protection relays. This leads to operation and, thus, line tripping during the build-up of the island [83].

(iii) ***The primary response from the black start unit***: Sometimes island fails to supply on separation from the grid as islanded unit fails to provide a primary response. Even though the island separated successfully from the main grid, there is a possibility that it may collapse within a few seconds. This signifies the benefits of larger interconnection and the need for primary response and generator AVR control. As there is no primary response from the generator, the operators can manually stabilise the island by controlling the governor. It may be due to the tripping of the unit because of low frequency. The generator selected as a potential black start source should be capable of operating in speed control mode during normal operation [116].

(iv) ***Secondary Collapse***: For the self-starting black start unit from the local DG set, BSU is started at a reduced terminal voltage than the rated one by keeping the function of generator control in manual mode. So that there is no secondary collapse when the island is built up due to high voltage [70].

(v) **Tripping of Units**: The island is synchronised with the rest of the grid successfully. The generator experienced a jerk immediately after check-synchronisation and tripped on excitation protection [118]. The unit is normalised afterwards. However, the generator occasionally jerked and tripped on excitation protection shortly after the check-synchronisation. The operators can identify and correct equipment issues due to such surprises regarding the facilities available for system restoration at significant substations. This underlines the necessity of routinely checking the readiness of the exercise's personnel and equipment. One extra unit is maintained on standby if the original unit trips [36].

(vi) ***Charging an EHV Line***: One of the primary challenges of this black start exercise was charging an EHV long transmission line from a single generator (isolated) to create an island. Last, synchronising the island with the power grid [64].

(vii) ***Re-striking across the breaker***: Just after opening the bus couplers, the corresponding isolators are open, so re-striking across the breaker contacts should be avoided. That is due to potential differences which can be developed across the contacts at the time of the black start drills [71].

(viii) ***Fluctuations in frequency***: The black start unit is started after changing the governor droop setting so it can operate over a wide frequency range. Therefore, it can accommodate fluctuations in frequency at the time of separation from the grid [68].

(ix) ***Ferranti effect***: Due to the Ferranti effect, the generator may start with a voltage somewhat lower than the rated until the load is added. The augmentation of load should be done in the smallest possible increments. The frequency should be somewhat higher than 50 Hz when adding block loads. This, along with the quick governor action, would ensure that the unit did not trip on a frequency that was too high [65].

(x) ***Size and Complexity***: Power systems are growing in size and complexity throughout the world. Therefore, conventional approaches used for PSR need to be continuously upgraded, tested and validated [33].

(xi) ***Black Start Resources***: In most Hydro Power Stations, DGs are used as BSRs Diesel generators have numerous drawbacks, particularly concerning the environment, such as high costs, the need for regular maintenance, noise pollution, susceptibility to corrosion, and the necessity of winterising the generator for use in cold weather. Additionally, diesel generators can be prohibitively expensive and are often under-utilised as backup power sources [32].

(xii) ***Energy storage:*** When using renewable energy resources as black start sources, Energy Storage (ES) devices must be employed. However, these ES devices have limitations in terms of charge/ discharge power and capacity. If the output of renewable energy sources is insufficient or fluctuates significantly during the black-start process, it can pose challenges and limitations. There is a possibility of overcharging and over-discharging of ES, resulting in black-start failure [120].

5.2. Asynchronous-based HGU (DFIM)

(i) ***Maintenance of the ' pre-charge" state of DC Link excitation:*** DFIM is practically unable to initiate a black start due to the inadequacy of any auxiliary source to maintain the ' pre-charge"

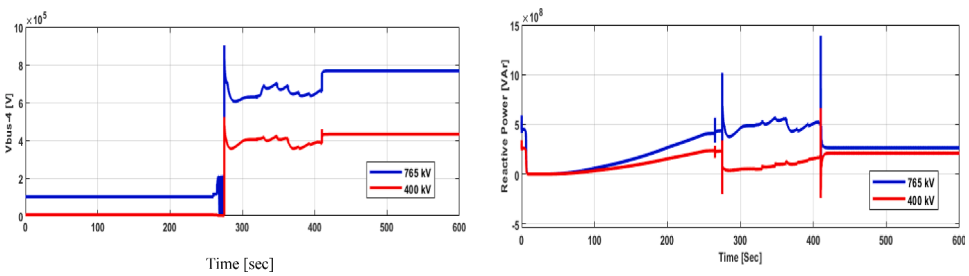


Fig. 16. Bus measurements for extra high voltage transmission line, 400 kV and 765 kV during black start exercise (a) Bus 4 voltage and (b) Reactive power.

state of DC Link. Say, e.g., Tehri PSP is there with DC-Link voltage of 2.5 kV. It requires a large-size DG set to pre-charge the DC-link itself [121,98].

(ii) ***DC-Link Stability:*** DC-link stability is not guaranteed with resources such as wind turbines & hydro turbine generators, which commonly use DFIM [122].

(iii) ***Impact of sub-synchronous oscillation (SSO) in view of blackout:*** Series compensation is given to the transmission line to improve power transfer capability, but the adverse effect is to create SSO. Subsynchronous oscillation creates a circulating current in DFIM converters that may create failure of the converters, which can lead to blackout [123].

5.3. Emerging threats

5.3.1. Cyber threats

Cyber exposure to control systems in industries has increased that was entirely disconnected from the outside world. This results in increased vulnerabilities to cyber attacks in the grid, which leads to increased risk to the grid operation. Also, there are chances of failure in restoring the system after the disturbances. Therefore while planning the black start exercise, critical control systems should be considered with due care to minimise cyber threat impacts [124].

5.3.2. Climate change

Climate change these days is imposing a threat to the infrastructure of the nation. Natural disasters such as hurricanes, thunderstorms, superstorms sandy, lightning strikes, blizzards, etc., have devastated parts of the power grid. The recovery from these incidents is expensive and time-consuming and needs dedicated approaches for addressing these natural disasters [125].

5.3.3. Geomagnetic disturbances (GMD)

Events such as GMDs also have an adverse impact on the transmission systems as the geomagnetically induced current can cause saturation of transformers.GMD caused the power outage in Quebec's system in 1989 and also Odenwalder 2009 [126].

5.4. Unconventional black start methods

Innovative black start and restoration techniques minimise the possibility of blackout damage. With the change in technologies and capabilities, new approaches are preferred to traditional methods. Different unconventional black start methods are available that are not so common but have proven successful in real-life events and can be used for a black start in future [53].

5.4.1. Grid-level and distributed batteries for black start

The grid can be stored by batteries as well. Currently, some stations are using combined cycle turbine generators, but we will be shifting to renewable energy sources. The batteries are not capable of providing energy for a long time. Batteries are not designed for this purpose but for keeping the grid stable while RES provide the required energy. Hence, we can shift renewable energy sources with battery energy storage systems for black start purposes. Research is going on using distributed energy resources (DER) in the distribution system for black start operations [120].

5.4.2. Voltage source converter

Conventional methods use line current converters that need a high amount of reactive support. Also not operate in the dead system due to commutation power requirements. Emerging technologies are using voltage source converters that include PWM-based VSC HVDC schemes that don't have such restrictions [127]. VSCs needs to be used for black start purpose. For this, we need station service power that batteries or DGs can give. VSCs are preferred because of relatively low power requirements.

5.5. Power failure challenges in developing nations

The major power failure issues faced by developing nations, encompassing outdated infrastructure, limited investment, fossil fuel dependence, inadequate access, theft, maintenance gaps, governance challenges, rapid urbanization, energy source diversity, climate vulnerability, financing constraints, skill shortages, and political instability [128]. The authors emphasize that addressing these challenges requires policy interventions, investments, capacity building, and international cooperation, with a focus on sustainable energy practices, renewables, infrastructure enhancement, and an enabling environment. They cite India's initiatives, such as "Power for All," renewable targets, and improved discom finances, as positive steps, while encouraging private sector participation, grid resilience, energy efficiency, and cleaner energy sources to combat power failures and energy challenges effectively.

5.6. Future prospects

5.6.1. Advanced control algorithms

Develop more advanced and adaptive control algorithms that can optimize the synchronization and operation of both synchronous and asynchronous hydro units during grid restoration. Machine learning and artificial intelligence techniques can be employed to enhance the decision-making process in real-time [33,96].

5.6.2. Hydro unit integration

Investigate the integration of various types of hydro units, including traditional synchronous generators, asynchronous generators, and even newer technologies like variable-speed hydro units. Evaluate their respective advantages and disadvantages in blackout restoration scenarios [129].

5.6.3. Grid resilience

Research how to make power grids more resilient to blackouts in the first place. This could involve the development of smart grid technologies, distributed energy resources, and microgrids that can help reduce the impact of disturbances [130].

5.6.4. Energy storage integration

Explore the integration of energy storage systems (e.g., batteries, supercapacitors) with hydro units to enhance their black-start capabilities. This can improve grid restoration speed and reliability [120].

5.6.5. Cybersecurity

Focus on enhancing the cybersecurity of power grid control systems to protect against cyber-attacks during blackouts and grid restoration efforts. Develop strategies to ensure the secure operation of hydro units in such scenarios [131].

5.6.6. Multi-Objective optimization

Consider multi-objective optimization techniques that balance factors like grid stability, economic efficiency, and environmental impact during the restoration process. This can help make more informed decisions during emergencies [42,96].

5.6.7. Policy and regulation

Analyze the regulatory and policy frameworks governing grid restoration procedures and recommend updates or improvements to facilitate the integration of advanced technologies and methodologies [132].

5.6.8. Interdisciplinary collaboration

Encourage collaboration between power systems engineers, control

systems experts, cybersecurity specialists, and data scientists to address the complex challenges associated with grid restoration comprehensively [133].

6. Conclusion

This paper has given an overview of the black start exercises applied to hydroelectric generating units and discussed several blackouts worldwide and their causes. Blackout is the worst type of emergency faced in integrated grid functioning. Power outages can result from various factors, such as delays in clearing faults, generator tripping, equipment malfunctions, human errors, and other contributing causes. Various black start methods have been adopted in different parts of the world. Tehri HPP is considered for the case study to illustrate the importance of technology and advanced controllers for providing black start services. This study shows that variable speed hydro is a new addition that can provide fast response services and be used for black start purposes. The findings reviewed in this paper will help system operators gain confidence in dealing with real-world blackout conditions while enhancing the performance of black start-capable generators. However, it needs more backup power to maintain the "pre-charge" state of the DC link. In most Hydro Power Stations, diesel generators are used as black start sources. However, diesel generators have many drawbacks, especially for the environment. Moreover, it is a very costly and highly under-utilised black start resource. Therefore, to reduce the diesel conception, the additional backup power requirement during black start can be fulfilled by renewable energy resources (RES) such as floating solar PV, wind, geothermal, etc., along with a battery energy storage system (BESS). At the same time, RES-BESS can be used to ramp up the power and support the renewable integrated grid system.

CRediT authorship contribution statement

Surabhi Verma: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Thanga Raj Chelliah:** Supervision, Funding acquisition, Project administration, Resources, Software.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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Data availability

The authors do not have permission to share data.

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